2. Dataset Description

This section outlines the process of hyperspectral data acquisition for wheat seeds and details the subsequent steps involved in preparing the raw data for analysis.

2a. Hyperspectral Image Acquisition

In the current research, we sourced the 50 popular wheat species harvested in 2022 from India's certified wheat growing institutes in Rajasthan, Karnataka, Madhya Pradesh, and Haryana. To shield these seeds from environmental factors, they were placed in plastic bags and maintained at a constant temperature of 4◦C in a refrigerator. The seeds were retrieved from refrigeration twenty-four hours before data collection and allowed to equilibrate to room temperature, which was maintained at (25 ± 1) ˚C with a relative humidity of 51 ± 5%. The moisture content of the wheat samples, assessed through the ISO 712:2009 oven drying method at temperatures between 130 to 140 ◦C, ranged from 12% to 14% on a wet basis.

Hyperspectral images of wheat seeds were acquired using a line-scan hyperspectral imaging system operating in reflectance mode. The complete imaging setup is illustrated in Fig. 1a. The camera was positioned 300 mm above the sample area, and both the camera and light source were powered on 45 minutes prior to image acquisition to ensure thermal stability and reduce spectral drift. To eliminate stray light interference, the entire HSI system was enclosed in a dark chamber and operated via a connected laptop. The HSI camera was equipped with a 2D photodetector array with a resolution of 320 × 168 pixels (spatial × spectral), enabling the capture of detailed spectral profiles. Wheat seeds were arranged in three circular clusters on a tray, as shown in Fig. 1b. For each variety, two hyperspectral images were captured by positioning the tray within the camera's field of view.

The acquired data were stored as a hyperspectral cube with dimensions p × q × λ, where p and q represent spatial coordinates, and λ denotes spectral bands. Due to the limited sensitivity of the InGaAs sensor at the spectral extremes, the first 14 and last 7 bands exhibited high noise and poor signal-to-noise ratios. These bands, corresponding to the wavelength ranges of 887.46–950.74 nm and 1694.04–1725.06 nm, were excluded from further analysis. The resulting effective spectral range was 955.62–1688.87 nm, comprising 147 distinct and reliable spectral bands. The software utilized for this purpose automatically calibrated the digital intensity values of the acquired hyperspectral images to ensure accuracy. Calibration involved the employment of both a dark reference and a white reference. The dark reference was obtained by covering the lens aperture with a cap, while a standard white fluorilon tile exhibiting a reflectance level exceeding 98% served as the source for the white reference. Subsequently, the software automatically computed the corrected reflectance intensities for the hyperspectral image using Equation 1:

$$I_C = \frac{I_{raw} - I_{dark}}{I_{white} - I_{dark}} \times 100 \qquad (1)$$

In this equation, I_c represents the corrected reflectance intensity value, I_{raw} signifies the reflectance intensity of the raw image, I_{dark} indicates the reflectance intensity of the dark reference image, and I_{white} represents the reflectance intensity of the white reference image.

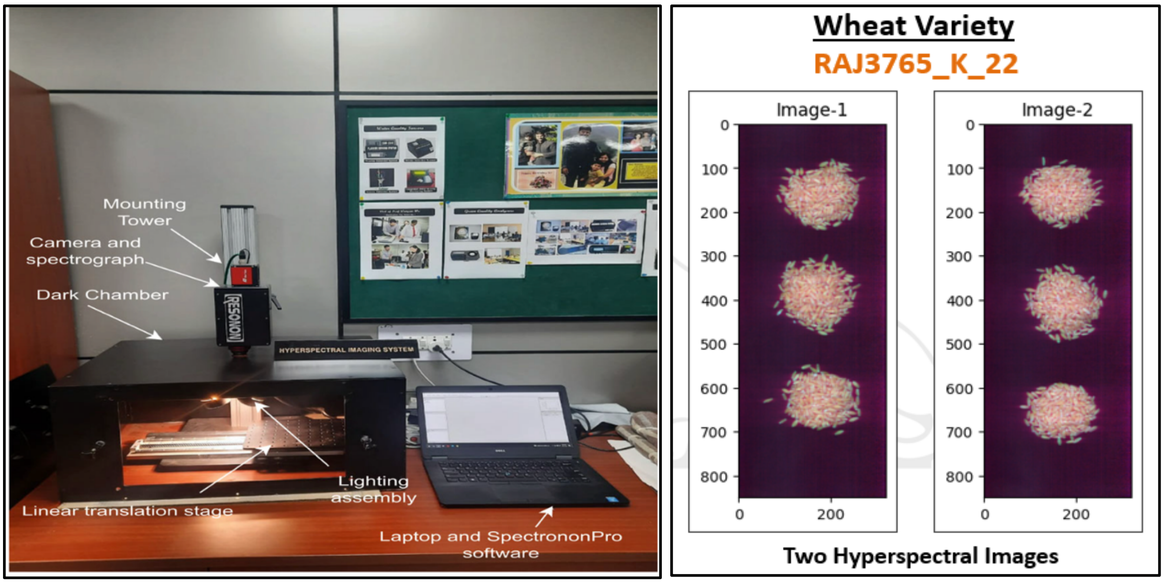

Fig. 1: (a) Hyperspectral Imaging System Set-Up (b) Two Hyperspectral Images (850 × 320 × 168) of RAJ3765_K_22 Wheat Variety

2b. Bulk Seeds Data Preparation

The dataset now comprises 100 hyperspectral images corresponding to 50 wheat seed varieties, each with dimensions of 850 × 320 × 147. The initial step in data preparation involves isolating the three circular seed regions from each hyperspectral image. This process began with the generation of a binary mask from the 60th spectral band using Otsu's thresholding method. To eliminate small gaps between adjacent seeds, a morphological closing operation was applied using a square structuring element of size 5. Following this, any remaining large holes within the connected regions were filled to ensure that all enclosed areas are solid, as illustrated in Fig. 2. The resulting binary image was then labeled, assigning unique identifiers to each connected component and key region properties such as height, width, and centroid coordinates were extracted for each labeled region. Using these properties, the boundaries of the seed regions were determined, and the three circular areas were isolated using cropping, as depicted in Fig. 3a.

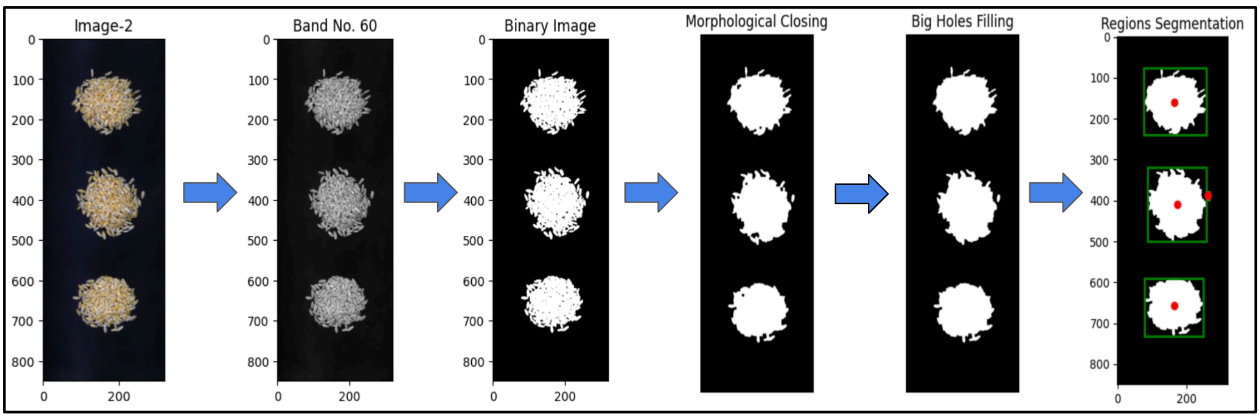

Fig. 2: Steps involved in Circular Regions Segmentation from each Hyperspectral Image

The next step involves extracting uniformly sized hyperspectral image patches from each circular seed region. A central patch of dimensions 56 × 56 × 147 was first cropped. The cropping window was then shifted by 18 pixels twice in each of the four cardinal directions -

North, East, South, and West - yielding eight additional patches. Likewise, the window was shifted twice by 11 pixels in each of the four diagonal directions to generate eight more patches, as illustrated in Fig. 3b. In total, 17 hyperspectral image patches of size 56 × 56 × 147 were extracted from each region, as shown in Fig. 3c. Despite some degree of overlap, these patches provide sufficient diversity and are well-suited for training deep learning models.

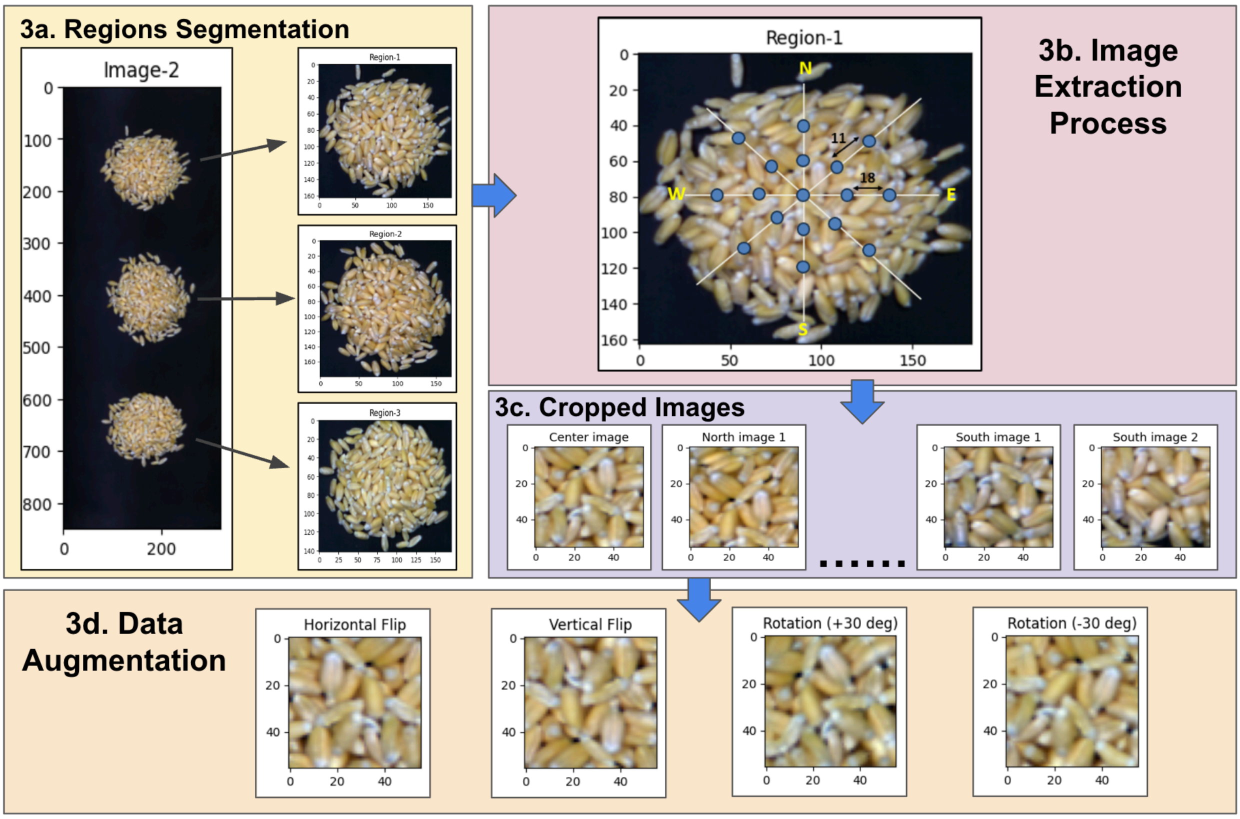

Fig. 3: (a) Extraction of individual regions from the hyperspectral image. (b) The process to extract 56 x 56 x 147 sized images from each circular region. (c) Cropped images from each region. (d) Augmented images

Finally, four data augmentation techniques namely horizontal flip, vertical flip, 30° anticlockwise rotation (+), and 30° clockwise rotation (-) were applied to each extracted hyperspectral image. As illustrated in Fig. 3d, this process generated four additional images per cropped image, resulting in 85 images from each circular region. Consequently, 510 images were generated for each seed variety, yielding a total of 25,500 images across all 50 varieties. For dataset partitioning, patches extracted from the upper two circular regions (i.e., four regions across both original hyperspectral images) were used for training, while the lower circular region from one image was used for validation and the other for testing. This resulted in 17,000 training images, 4,250 validation images, and 4,250 test images.